

## Modelo de ecuaciones integro-diferenciales

M1  $V_e(t) = R_1 i_1(t) + \frac{1}{C} \int (i_1(t) - i_2(t)) dt$

M2  $0 = \frac{1}{C} \int (i_2(t) - i_1(t)) dt + L \frac{di_2(t)}{dt} + R_2 i_2(t) \rightarrow \frac{1}{C} \int (i_1(t) - i_2(t)) dt = L \frac{di_2(t)}{dt} + R_2 i_2(t)$

Salida  $V_S(t) = R_2 i_2(t)$

## Transformada de Laplace

M1  $V_e(s) = R_1 i_1(s) + \frac{1}{Cs} (i_1(s) - i_2(s))$

M2  $\frac{1}{Cs} (i_1(s) - i_2(s)) = (Ls + R_2) i_2(s)$

$i_1(s) - i_2(s) = Cs(Ls + R_2) i_2(s)$

$i_1(s) = (LCs^2 + CR_2s + 1) i_2(s)$

## Sustituir en malla 1

$V_e(s) = R_1(LCs^2 + CR_2s + 1) i_2(s) + (Ls + R_2) i_2(s)$

$V_e(s) = [(CLR_1)s^2 + (CR_1R_2 + L)s + (R_1 + R_2)] i_2(s)$

$V_S(s) = R_2 i_2(s) \rightarrow i_2(s) = \frac{V_S(s)}{R_2}$

$V_e(s) = \frac{(CLR_1)s^2 + (CR_1R_2 + L)s + (R_1 + R_2)}{R_2} V_S(s)$

## Función de Transferencia

$\frac{V_S(s)}{V_e(s)} = \frac{R_2}{(CLR_1)s^2 + (CR_1R_2 + L)s + (R_1 + R_2)}$

## Error en estado estacionario

$E(s) = V_e(s) - V_S(s)$   
 $= V_e(s) [1 - FT]$

$E(s) = \frac{1}{s} (1 - FT)$

$e_{ss} = \lim_{s \rightarrow 0} E(s)$

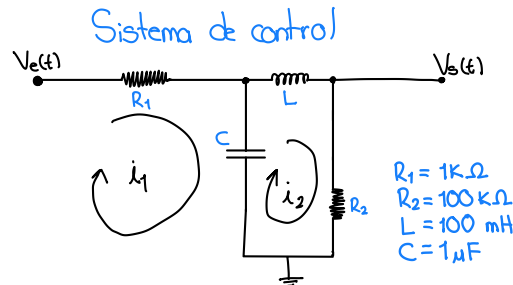
$e_{ss} = \lim_{s \rightarrow 0} \left( \frac{R_2}{(CLR_1)s^2 + (CR_1R_2 + L)s + (R_1 + R_2)} \right)$

$e_{ss} = \frac{R_1}{R_1 + R_2}$

$e_{ss \text{ control}} = \frac{1K}{1K + 100K} = 0.0099$

$e_{ss \text{ caso}} = \frac{1K}{1K + 1K} = 0.5$

## Estabilidad del Sistema



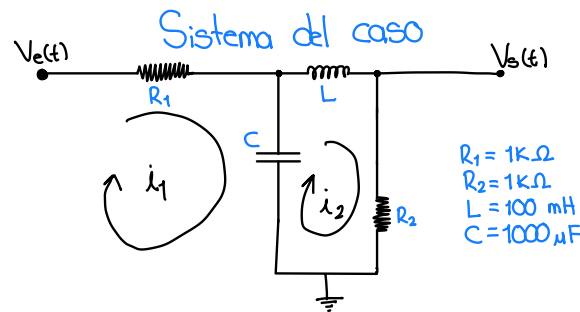
$$e_s = \frac{100 \times 10^3}{(1 \times 10^{-6} \cdot 100 \times 10^3 \cdot 1 \times 10^3)s^2 + ((1 \times 10^{-6} \cdot 1 \times 10^3 \cdot 100 \times 10^3) + 1 \times 10^3)s + (1 \times 10^3 + 100 \times 10^3)}$$

$$= \frac{1}{10000}s^2 + \frac{100001}{1000}s + 101000$$

$$S_1 = -1011.012035$$

$$S_2 = -998998.988$$

Sistema estable con una respuesta Sobreamortiguada



$$e_s = \frac{1 \times 10^3}{(1000 \times 10^{-6} \cdot 100 \times 10^3 \cdot 1 \times 10^3)s^2 + ((1000 \times 10^{-6} \cdot 1 \times 10^3 \cdot 1 \times 10^3) + 100 \times 10^3)s + (1 \times 10^3 + 1 \times 10^3)}$$

$$= \frac{1}{10}s^2 + \frac{10001}{10}s + 1000000$$

$$S_1 = -2.000200006$$

$$S_2 = -9998.9998$$

Sistema estable con una respuesta Sobreamortiguada